25901872 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Skill Development Components (SDC) - MG2SDCSEG100 - ANALOG AND DIGITAL AUDIO

(2025 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. What subjective term is often used to describe the sound of analog saturation? [K] / [CO1]
2. What produces the magnetic pattern on a recording tape? [K] / [CO1]
3. Identify the material commonly used in the manufacture of modern vinyl records. [K] / [CO1]
4. Define “Stop Edit” in an analog tape recorder. [K] / [CO2]
5. What is the output signal of a playback head? [K] / [CO2]
6. Define 'Print-through'. [K] / [CO2]
7. Name the circuit that holds the voltage steady for the ADC. [K] / [CO3]
8. State the theoretical dynamic range of 16-bit audio. [U] / [CO3]
9. Define Pulse Code Modulation (PCM) and state whether it is a compressed or uncompressed digital audio format. [K] / [CO3]

10. Expand the acronym MADI. [K] / [CO4]

11. Which operating system originally used AIFF as its native format? [K] / [CO4]

12. What does 0 dBFS represent in digital audio? [K] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain the typical signal flow from a microphone to a recording medium. [K] / [CO1]

14. Explain the impact of solid-state storage on modern recording technology. [K] / [CO1]

15. Explain how wow and flutter are simulated in modern tape plugins. [U] / [CO2]

16. What is modulation noise and how does it differ from standard hiss? [K] / [CO2]

17. Convert the decimal numbers 256, 512 and 1024 into binary form, showing the steps involved. [U] / [CO3]

18. How are logic gates applied in digital audio switching or routing? [U] / [CO3]

19. Define quantization error and explain how it leads to distortion. [U] / [CO3]

20. How does a master clock distribute timing information to slave devices? [U] / [CO4]

21. Explain how FLAC reduces file size without losing audio quality. [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

22. Compare the applications of portable digital recorders and professional studio recording systems in different environments. [K] / [CO1]

23. Describe essential maintenance routines for a professional tape machine, including cleaning and degaussing. [U] / [CO2]

24. Discuss the relationship between sampling rate and frequency range, including benefits of higher sample rate [U] / [CO3]

[10x2 = 20]